

3.5.1 Sequence

سیکوئنس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔

CS XI

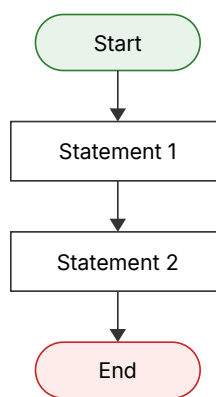
Chapter 3: Programming Fundamentals

Lecture 09 of 13

3.5.1 Sequence

LEARN IT

Sequence means statements are executed **one after another, strictly from top to bottom**, in the exact order they are written. This is the **default behaviour** of Python. No statement is skipped or repeated.



Sequence: control flows straight down from one statement to the next.

WORKED EXAMPLE

Input → Process → Output in sequence.

```

print("Enter two numbers")
a = int(input("Enter first number: ")) # Step 1: Input
b = int(input("Enter second number: ")) # Step 2: Input
total = a + b # Step 3: Process
print("Sum =", total) # Step 4: Output
  
```

Sample run (user types 7 and 5):

```

OUTPUT
Enter two numbers
Enter first number: 7
Enter second number: 5
Sum = 12
  
```

Each line runs exactly once, in order. If you moved line 4 (`total = a + b`) above the input lines, Python would stop with a `NameError` because `a` and `b` would not exist yet.

REMEMBER IT

Sequence is like reading a page: line 1, then line 2, then line 3. Most programs follow the pattern **Input → Process → Output**.

COMMON MISTAKE

Order matters. A variable must be given a value **before** it is used. Using it first causes `NameError: name '...' is not defined`.

اردو میں سمجھیں

سیکوانس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔ یہ پائتھن کا عام طریقہ ہے۔ کسی ویری ایبل کو استعمال کرنے سے پہلے اس میں ویلیو رکھنا ضروری ہے۔

CHECK YOURSELF

1. Predict the output:

```
x = 4
y = x * 2
x = 10
print(x, y)
```

2. What is the default control structure in Python?

Answers: 1. 10 8 — y was calculated when x was still 4; changing x later does not change y. 2. Sequence.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. In sequence, line 2 runs:

- a. After line 1 finishes
- b. Before line 1
- c. In parallel always
- d. Never

MCQ 2. `x=2; x=x+1; print(x)` prints:

- a. 3
- b. 2
- c. 1
- d. x

MCQ 3. Sequence bugs are often:

- a. Wrong order of statements
- b. Missing NAND
- c. OSI layer skip
- d. Pie explode

MCQ 4. You cannot print a variable:

- a. Before you assign it
- b. After you assign it
- c. Using print
- d. In comments

MCQ 5. Sequence is the default in:

- a. Python
- b. Only Prolog
- c. Only SQL joins
- d. Only CSS

MCQ 6. Swap two variables needs:

- a. A temp (or Python `a,b=b,a`) in the right order
- b. A K-map
- c. A router
- d. A sprint

Extra short questions

1. Define sequence.
2. Show a 4-step sequence to compute area of a rectangle.
3. What is a NameError from order?
4. Can sequence include comments?

Extra long question

1. Write two versions of a “read marks, total, average” program — one with statements in a sensible sequence and one in a broken order. Explain the error.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Statements executed in the order they are written.
2. Read L; read W; `area=L*W`; print area.
3. Printing x before `x = ...`
4. Yes; comments are not executed but sit in order in the file.

Long

1. Broken: print average before computing it, or total before reading marks. NameError or wrong value. Correct numbered steps.